these tools into the lab environment, with support for taking readings, making observations, plotting graphs, etc.

OLabs has been designed to address these concerns. First of all, there is a single portal under which all labs are accessible, irrespective of the subjects or the developer. See Figure 1, which shows the opening page of the website *www.olabs.edu.in*. It is fully browser based, and hence no local software or installations are required.

Figure 1: Home page of the OLabs

Generally, all the labs follow a common structure, as shown in Figure 2.

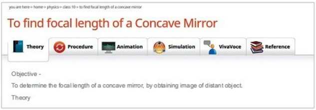

Figure 2: Structure of a virtual lab

There is a *Theory* tab, which documents the theoretical concepts addressed by the lab. The explanation is brief, since this is usually available in detail in the textbooks. This will help to check on the relevant formula, definitions, etc, when students are working on a lab, or as a precursor to performing the lab activity. The *Procedure* tab outlines the activity, step by step, both when you do the activity in a real lab, and when you do it using OLabs. This is usually the recommended procedure as per the curriculum, and OLabs reflects this sequence as far as possible.

The third tab is either a video, showing the lab activity in real life (e.g., in case of biology), or a teacher-guided view of the lab, which can be used to explain the lab to a class. The fourth tab is the actual lab, and offers an interactive platform for the student to perform the activity and control the relevant parameters within the intended scope of the lab activity. Where required, an observation window is provided to record the readings from different trials, and a graph plot depicting the same, visually. Where measurements are needed, a virtual scale is also provided.

A *Viva voce* tab is also provided with a few multiple choice questions for self-testing the concepts. The *Reference* tab includes some reading material, Web links, etc, that are relevant to the particular activity and can be used for further reading. This structure is one of the key USPs of OLabs, as it provides a complete ecosystem for directly adopting the system into the curriculum.

The labs follow the structure and process as outlined in the CBSE (Central Board of Secondary Education) lab manuals, further easing the adoption process. Despite this compliance to CBSE, adopting OLabs for state boards should not be difficult, since most of the syllabus and labs are expected to be common. Language can be a deterrent, and OLabs, in its current form, supports this too. The team behind the project has already translated OLabs into Hindi, Malayalam and Marathi. The same process can be extended to other languages—only translation of the text strings needs to be done. If readers can help to translate OLabs into any of the other Indian languages, please do get in touch with us.

Using OLabs

Given the approach we have followed, OLabs can be used in a variety of ways. Teachers can use the online labs in the classroom to explain the lab activity to be performed in the real lab. This helps to remove the surprise element when the student gets to the lab, and makes the lab session more effective. Teachers can also frame review questions with the lab as the backdrop, asking the students to predict what will happen under particular conditions. Students can use the lab before the physical lab session, to familiarise themselves with the process, and to chart out the expectations. As a post-lab activity, they can use it with more variations, to reinforce the concepts, and to answer any questions they may have. We are sure that creative teachers and students can come up with many more innovative uses for this resource.

The math lab

While the notion of a lab for physics, chemistry and biology is well understood, usually there are no formal labs for mathematics. The math lab in OLabs is envisaged as activities to reinforce conceptual relationships—e.g., the formula for the volume of a sphere. In the curriculum, these are called activities, and usually involve using resources such as paper, glue, sand, thread, scissors, etc, and making shapes and demonstrating similarities. These activities are captured into an interactive online experience allowing the student to perform these virtually. Students can, thus, draw a variety of triangles and see how a particular formula or theorem holds true for each of them. As in real life, they have choices when selecting a point, which step to do in which order, and so on. The system ensures that wrong steps are not performed.